

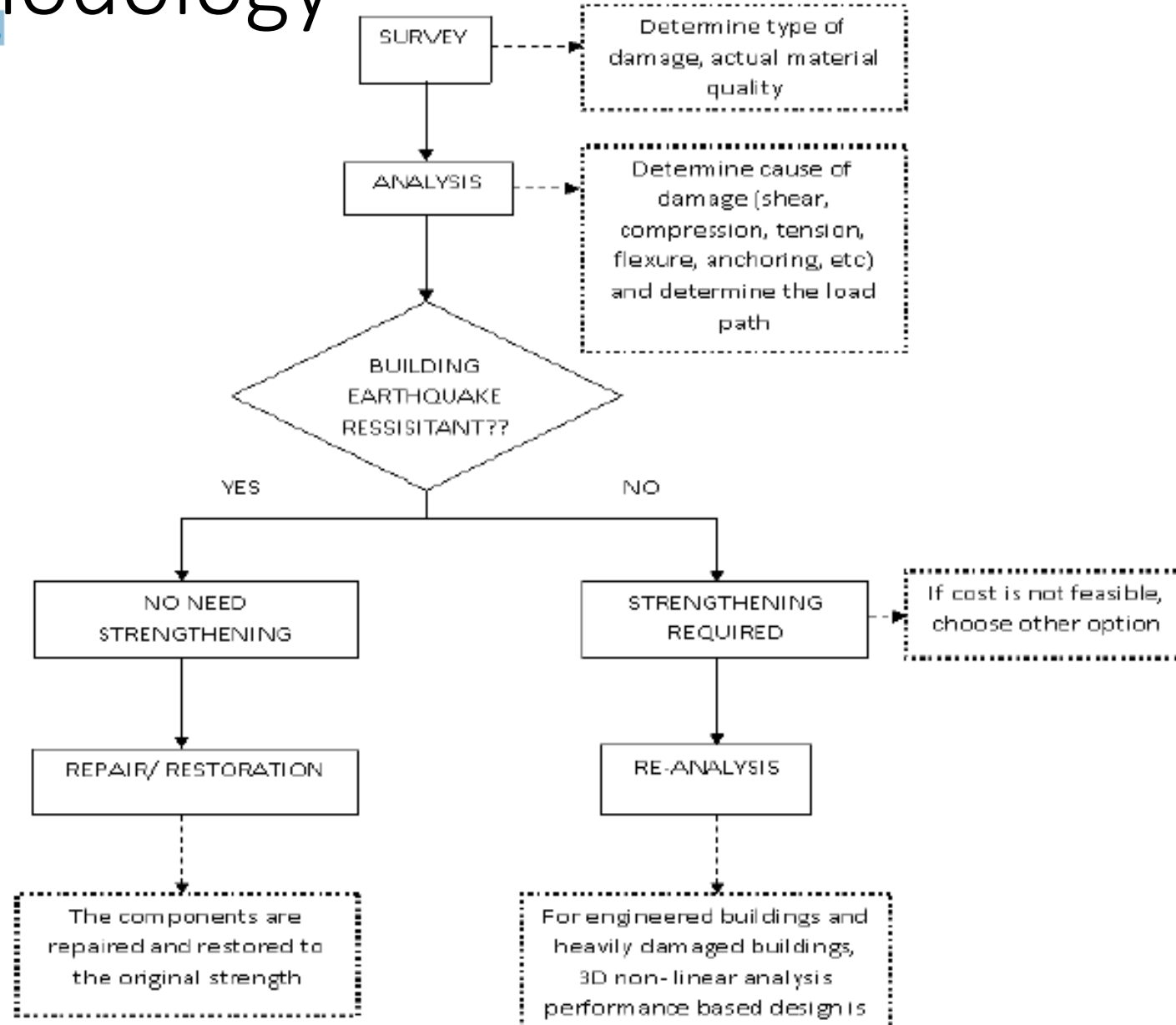
Informal Discussion on Retrofitting

CLPIU, Babarmahal

Concept of Repair, Restore and Retrofit

- Repair: to resume all its previous functions, Does not improve structural strength
- Restore: restore the structural strength prior of Damage
- Retrofit: Improvement over original strength

Methodology



Retrofitting
(Seismic
Strengthening)

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graph TD; A[Retrofitting (Seismic Strengthening)] --> B[For Adobe Building (Low Strength Masonry)]; A --> C[For Masonry Structures]; A --> D[For RCC Structures];
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For Adobe Building
(Low Strength
Masonry)

For Masonry
Structures

For RCC Structures

Retrofitting of Adobe Structure

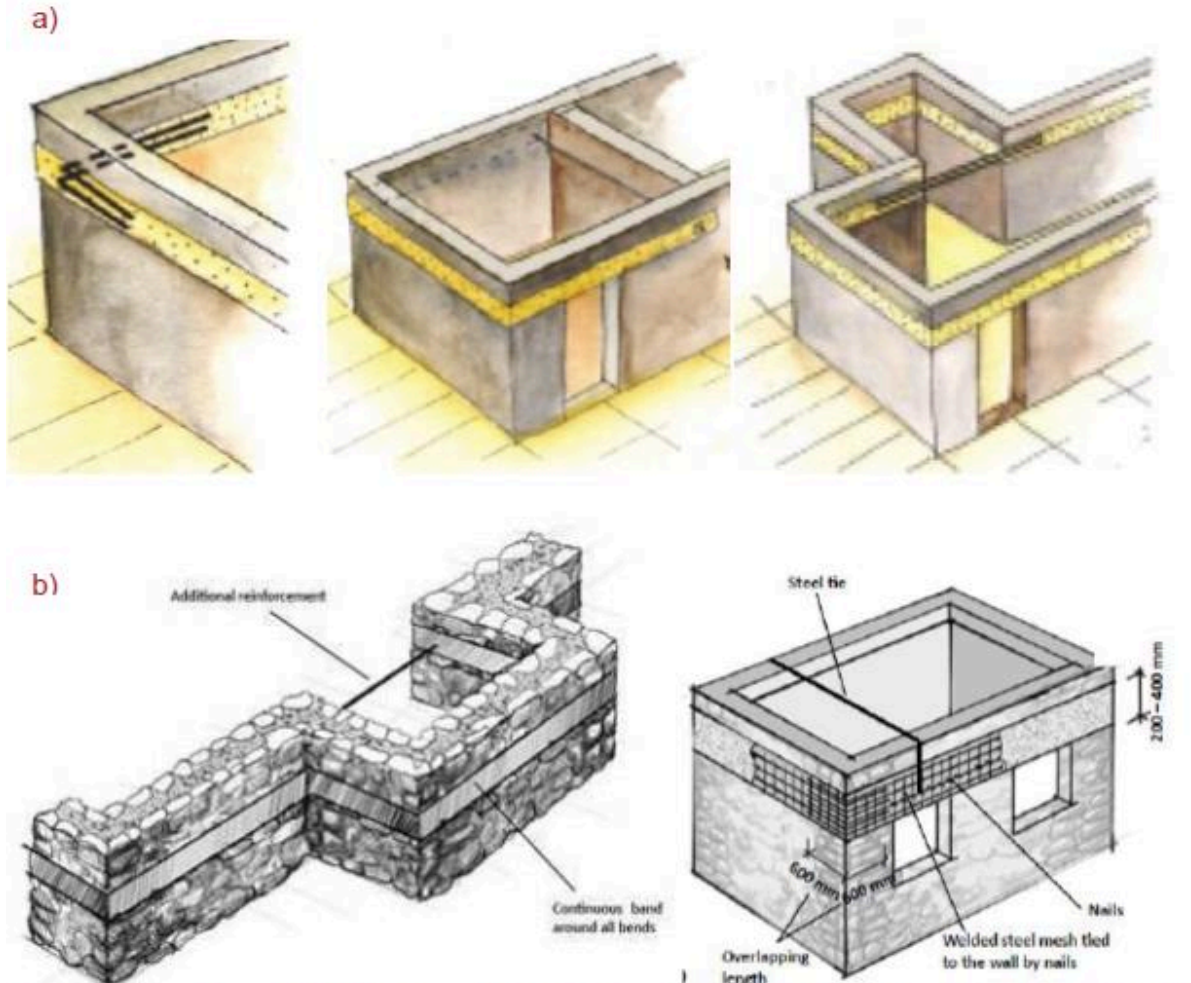


Figure 4-6 Gable band

Figure 4-5a) Seismic belts around various locations (source: UNDP, UNECO & GOI, 2007) b) Additional reinforcement (source: R. Desai and GOM 1998)

...Retrofitting of Adobe Structure

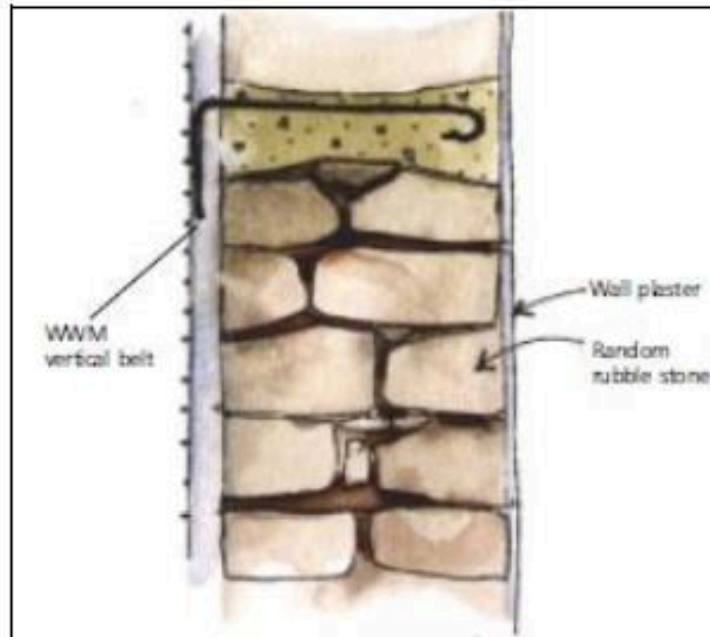


Figure 4-10 Anchoring WWM to random rubble wall with shear Connector (source: UNDP, UNECO & GOI, 2007)

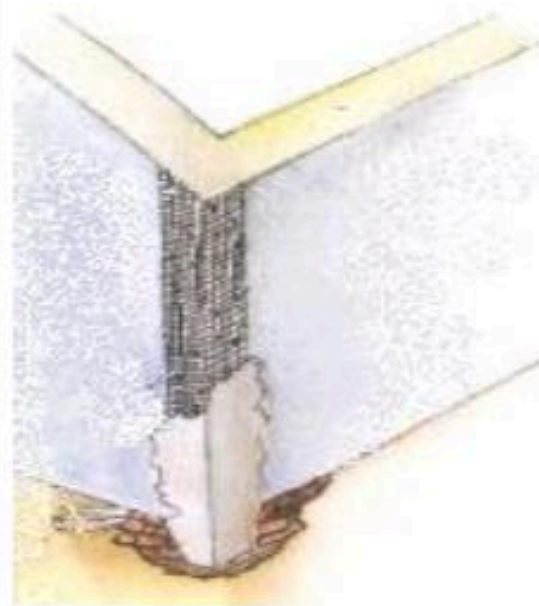
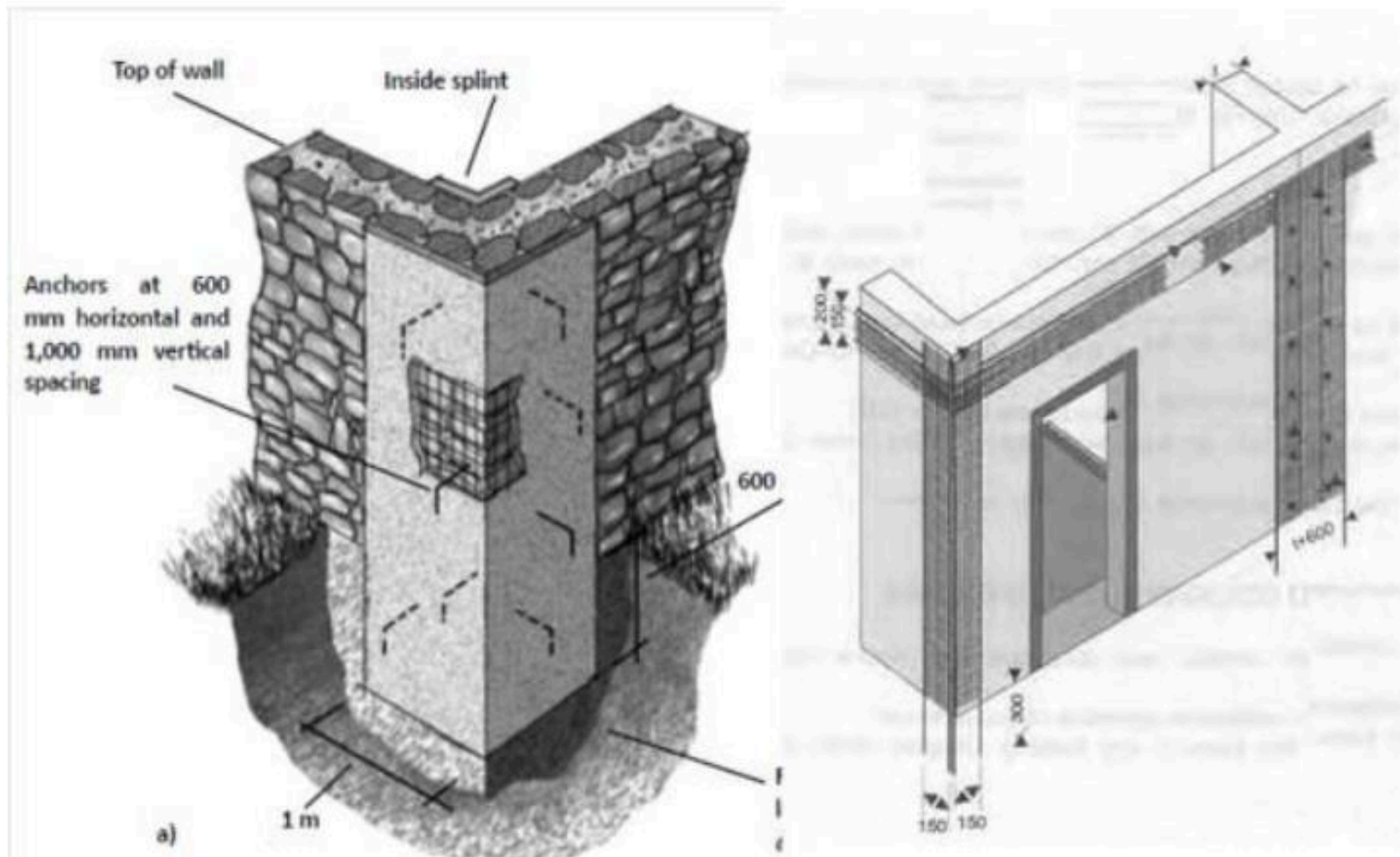


Figure 4-11 Plastering vertical WWM belt using cement plaster (source: UNDP, UNECO & GOI, 2007)

...Retrofitting of Adobe Structure



...Retrofitting of Adobe Structure



Figure 4-13: Encasing around window and door openings(source: UNDP, UNECO & GOI, 2007)



Figure 4-14 Full scale adobe model reinforced with pp band after a shake table test



Figure 4-15PP band mesh (source: Megura and Mayorca)



Figure 4-16 PP band retrofitted house before mortar laying (source: Megura and Mayorca)

... Retrofitting with Bamboo



Retrofitting for Masonry Buildings

Steel Wire Meshing/ Ferro cement Plating

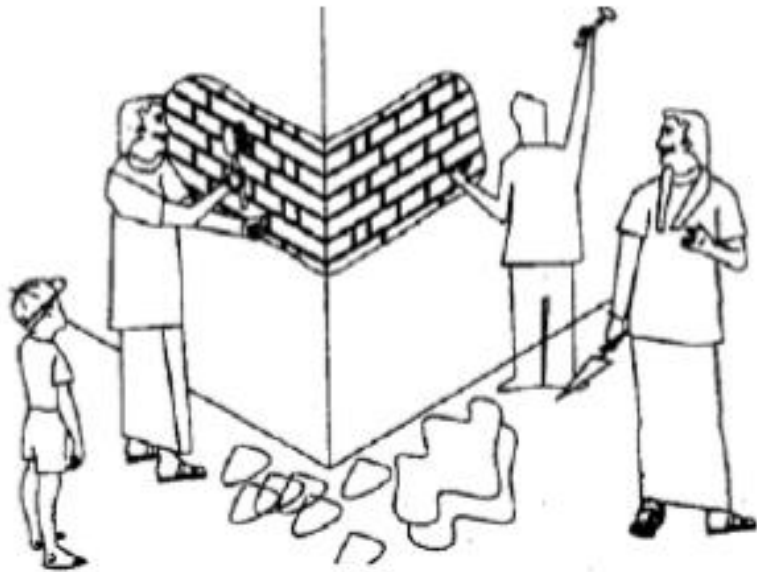


Fig.1:- Removal of Plaster & Raking of Joints

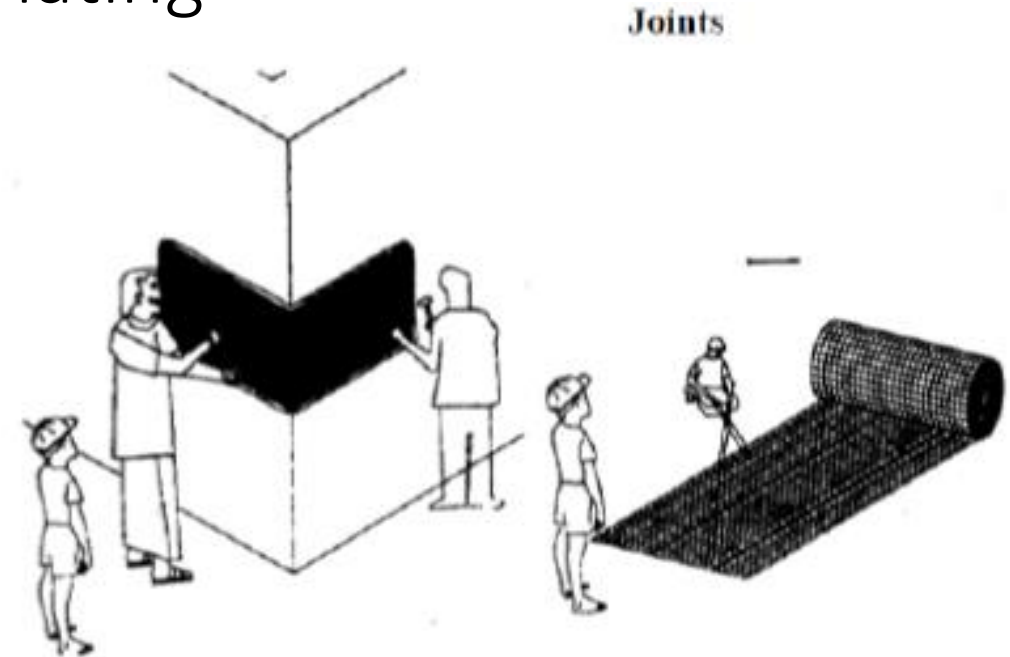
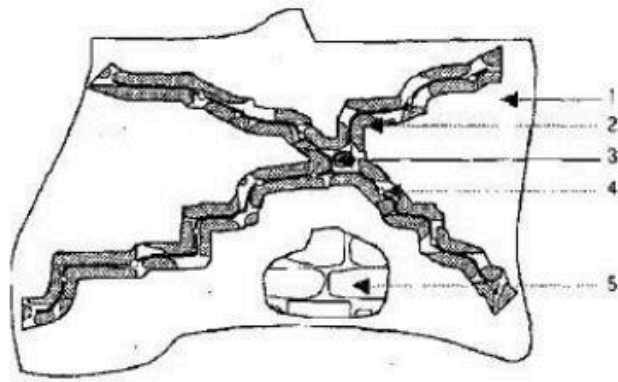


Fig. 2 :- Galvanised steel wire mesh

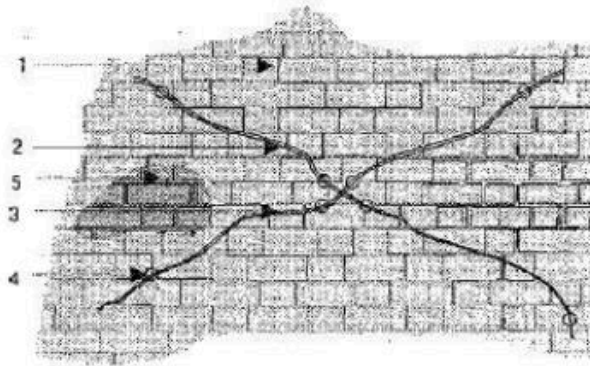
... Retrofitting Masonry..

Grouting

Step-1 Remove the plaster in the vicinity of crack exposing the cracked bare masonry



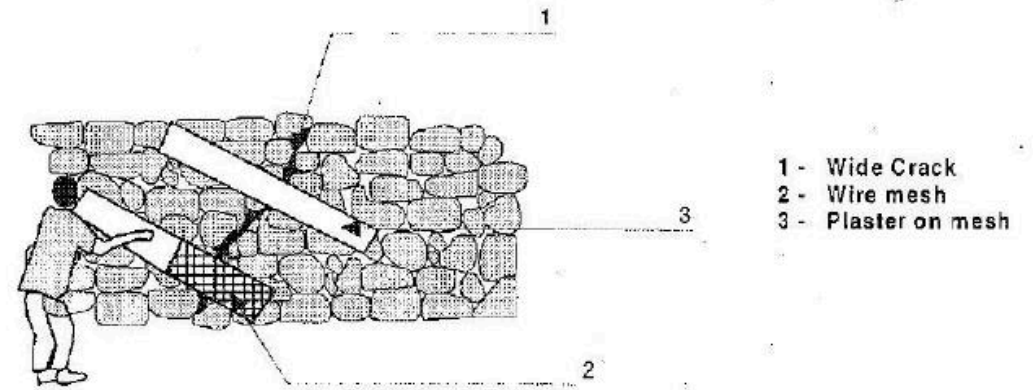
(a) Stone wall



(b) Brick wall

- 1- Plaster
- 2- Plaster removed and cracks cleaned
- 3- Cracks sealed with mortar or putty
- 4- Grout ports
- 5- Plaster fallen to be done again.

Figure 10-1 Filling grout in cracks



- 1 - Wide Crack
- 2 - Wire mesh
- 3 - Plaster on mesh

Figure 10-2 Fixing mesh across wide cracks

... Retrofitting Masonry.. Jacketing

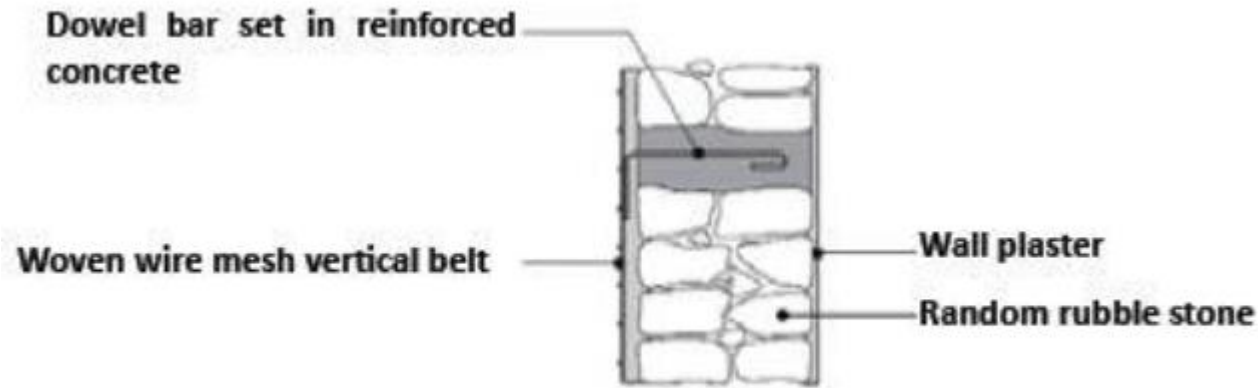


Figure 10-4 Single-sided jacketing showing steel dowels

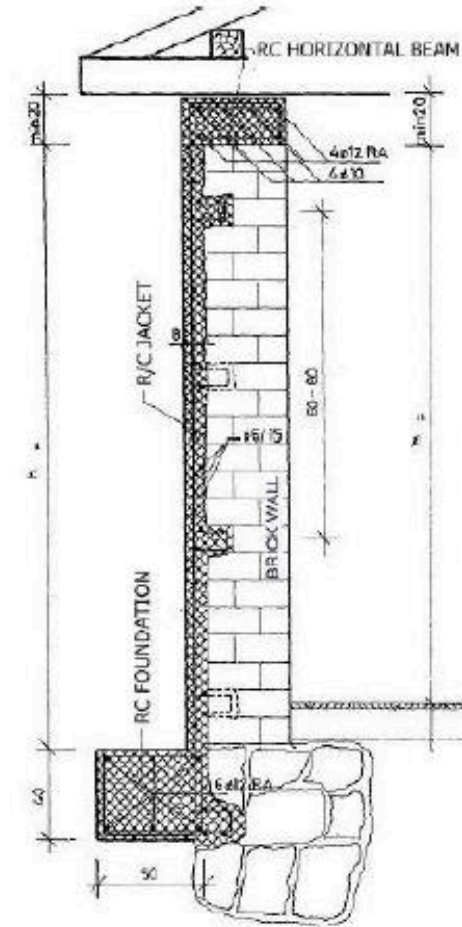


Figure 10-3 Application of RC coating

... Retrofitting Masonry..

Vertical Reinforcement

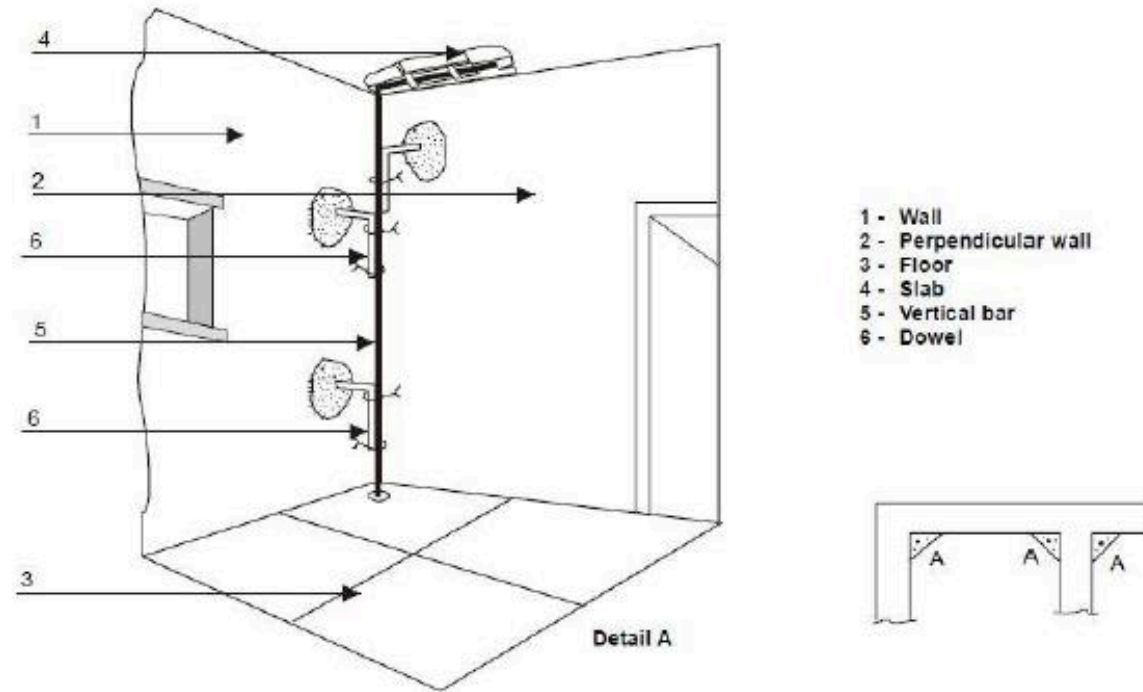
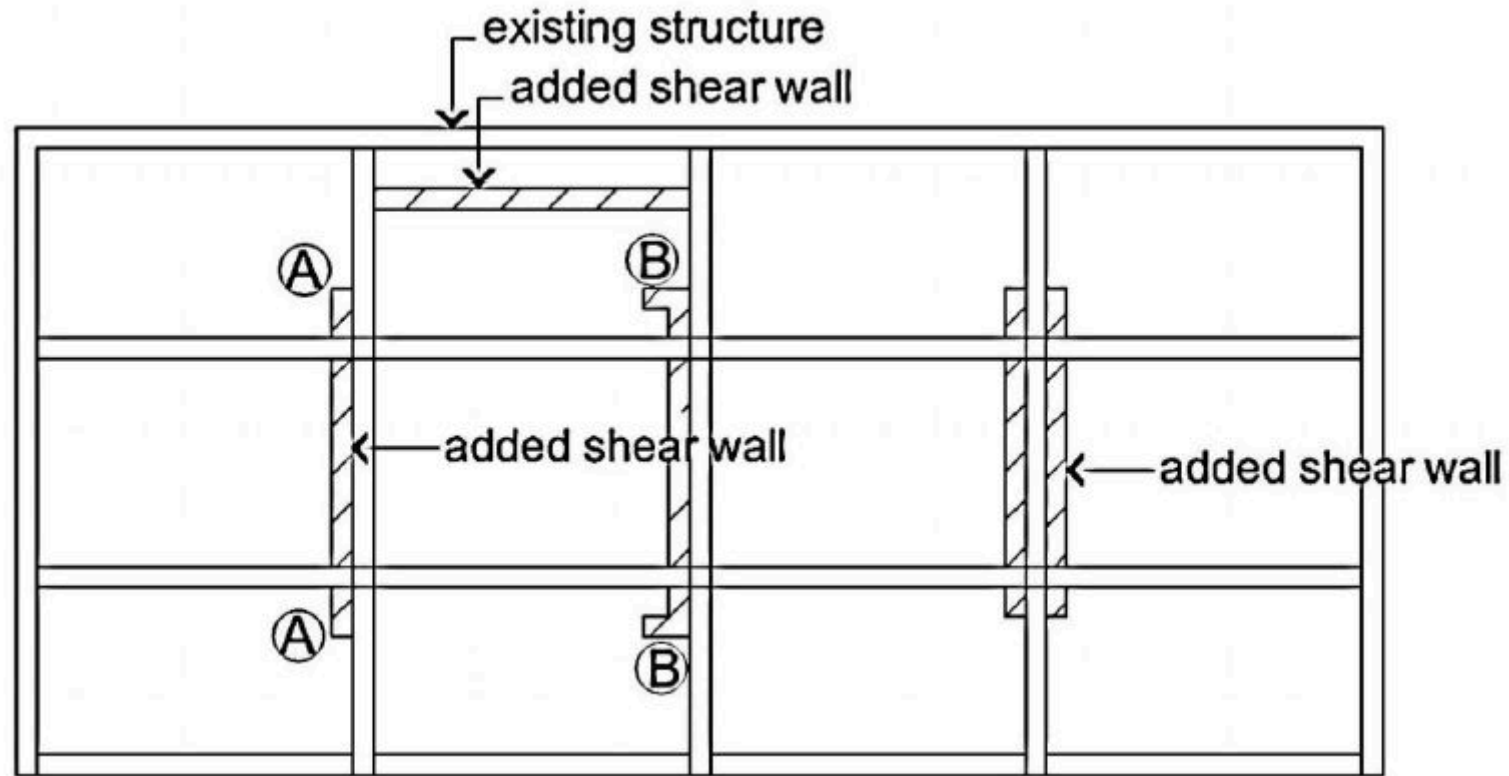


Figure 10-13 Vertical Reinforcement at corner and Junctions of Walls

Retrofitting RCC Structures

Adding Shear Wall



...Retrofitting RCC Structures

Eg. Adding Buttress Wall



...Retrofitting RCC Structures

Truss and Bracing

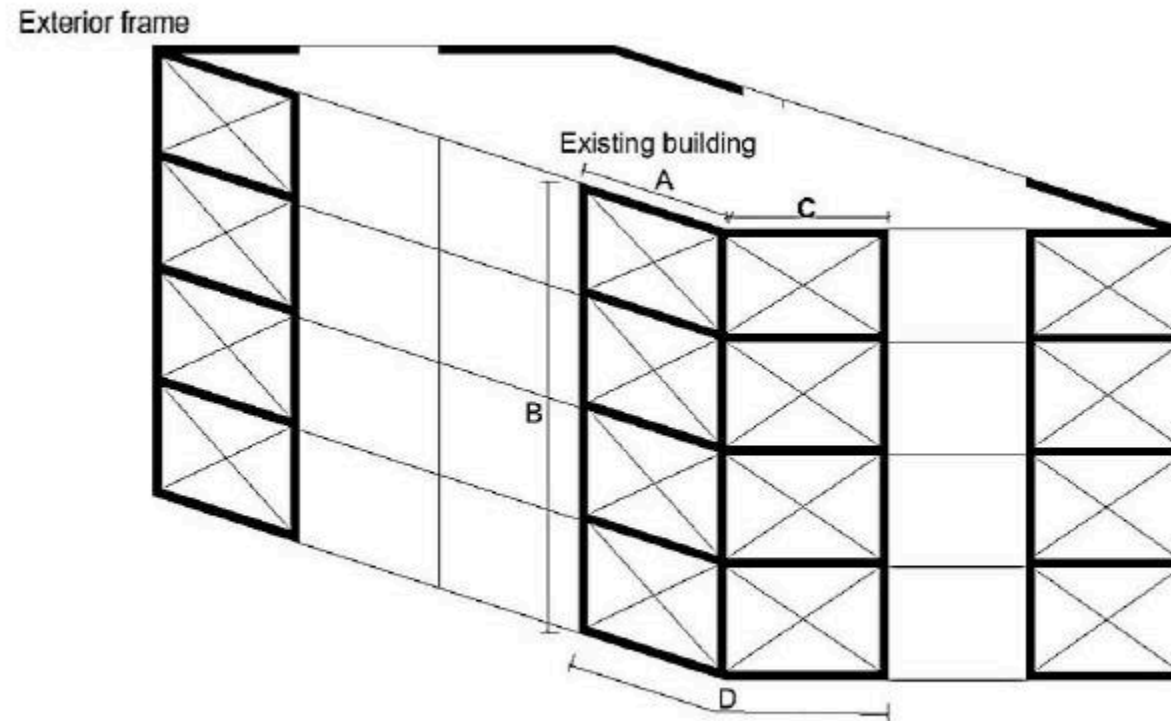


Figure 6-4 Exterior frame (steel framed brace)

...Retrofitting RCC Structures

Strengthening the preexisting elements



Roughening of old surface by chipping of existing concrete surface



Anchorage of new reinforcement with the existing structure

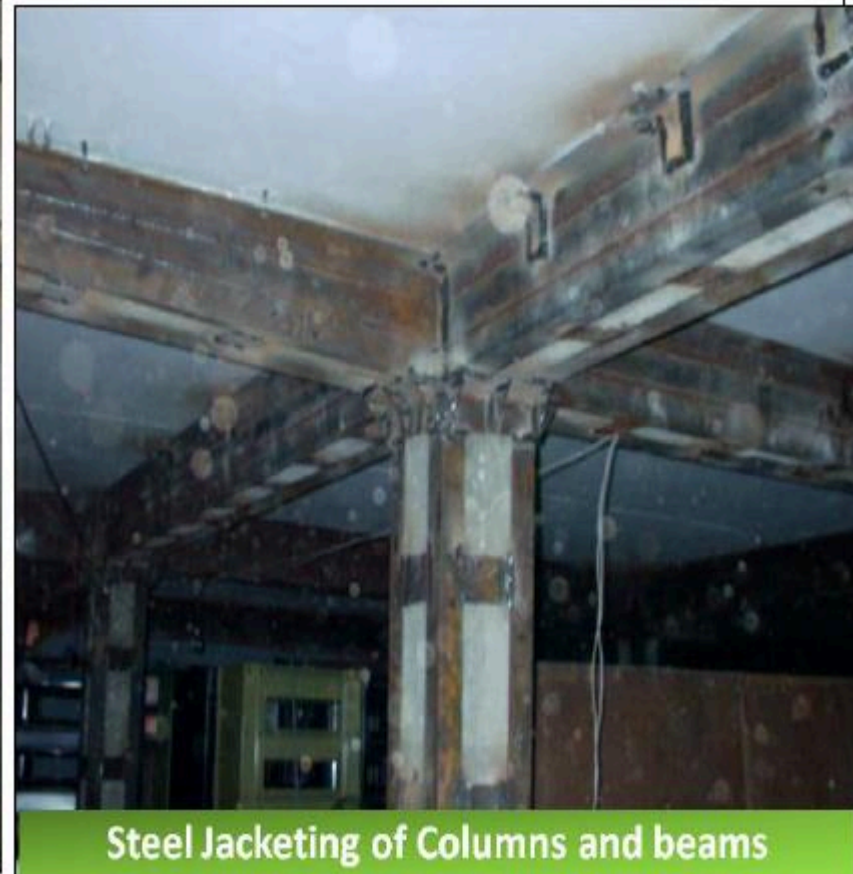
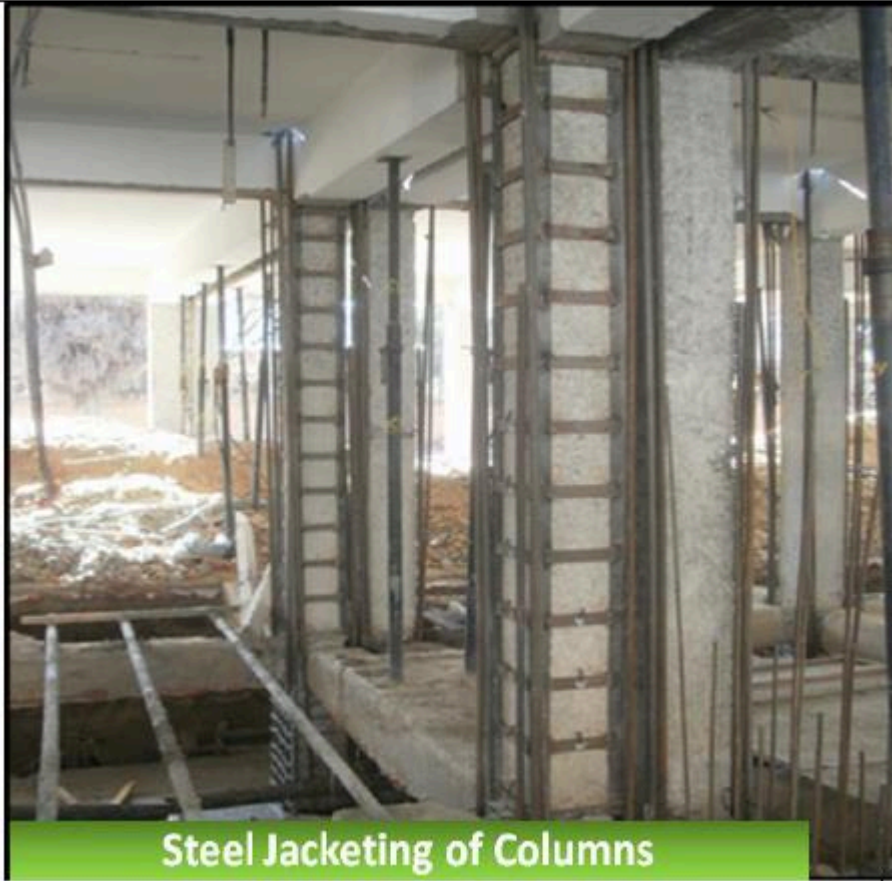
...Retrofitting RCC Structures

RC Jacketing of Columns



...Retrofitting RCC Structures

Steel Jacketing of Columns



...Retrofitting RCC Structures

Beam Jacketing



...Retrofitting RCC Structures Similarly



Reinforcement layout at foundation for retrofit



Shear wall addition with column jacketing



Shear wall addition with column jacketing

Others

- Base Isolation
- Dampers

Content based on

- Seismic Retrofitting Guidelines of Buildings in Nepal
 - Adobe Building
 - Masonry Structures
 - RCC Structure
- Government of Nepal

Summary

- Overview of Retrofitting solutions discussed internally
- Status of Official Documents as per to Retrofitting
- The types of Buildings covered broadly